

Physical-Organic Insights into Terminal Alkyne Acidity: Decoupling Electronic Substituent Effects via Semi-Empirical Quantum Descriptors and Interpretable Linear Models

Aniema Ekemini Johnson^{*,†} and Ivara Samuel[‡]

[†]*Independent Researcher, Calabar, Cross River State, Nigeria*

[‡]*University of Cross River State, Calabar, Cross River State, Nigeria*

E-mail: ekemjah5@gmail.com

Abstract

Predicting the *sp* C–H acidity (pK_a) of terminal alkynes is central to understanding homogeneous catalysis mechanisms, synthetic couplings, and molecular interactions in medicinal chemistry. Traditional high-level Density Functional Theory (DFT) calculations capture these values with high fidelity but incur heavy computational costs that obscure direct physical interpretation over broad structural libraries. In this work, we present a physical-organic framework that deconvolutes the structural and electronic features controlling terminal alkyne pK_a values by coupling semi-empirical tight-binding quantum mechanics (GFN2-xTB) with an interpretable, regularized Ridge Regression architecture. Our model demonstrates exceptional predictive agreement with experimental reference scales ($R^2 = 0.8611$, RMSE = 1.17 pK_a units) while running under

3.0 s per molecule. More importantly, feature weight analysis offers a direct, quantifiable validation of frontier molecular orbital theory: localized LUMO stabilization serves as the dominant thermodynamic driver controlling downstream acetylide anion stability. By profiling diverse virtual chemical matrices, including highly electron-withdrawing ethynyltosyl variants ($pK_a = 24.12$) and sterically hindered alkyl-substituted systems ($pK_a = 31.61$), this framework demonstrates how machine learning can be utilized not merely as a predictive black box, but as a transparent interpretive lens for core physical chemistry phenomena.

Introduction

Terminal alkynes serve as vital building blocks across molecular electronics, physical organic chemistry, and chemical biology. The properties of the *sp*-hybridized C–H bond uniquely govern the mechanical rate and selectivity of foundational synthetic transformations, such as Copper(I)-Catalyzed Azide-Alkyne Cycloadditions (CuAAC) and Sonogashira cross-couplings. Thermodynamically, these reactions are directly driven by the local acidity (pK_a) of the alkyne proton and the corresponding structural stability of the conjugate acetylide base. In a broader context, fine-tuning alkyne proton dissociation parameters represents a vital tactical lever for altering local molecular dipoles, optimizing electrostatic non-covalent interactions, and altering molecular reactivity profiles.

Quantifying these substituent-driven electronic variations over diverse molecular spaces historically relies on high-level hybrid functional Density Functional Theory (DFT) calculations. While DFT provides precise calculations of free energies, it functions primarily as an endpoint evaluation metric rather than an explanatory mechanism for rapid structural design rules. Furthermore, the steep scaling cost of *ab initio* methods complicates high-throughput electronic exploration. Conversely, while standard deep learning cheminformatics pipelines bypass this bottleneck using topological fingerprints, they obscure the underlying physical-organic principles, acting as uninterpretable "black boxes" that hide the causal link between

electronic structures and chemical reactivities.

In this work, we address this challenge by leveraging semi-empirical extended tight-binding theory (GFN2-xTB) alongside an explainable linear ridge regression framework to explicitly map the physical relationships dictating terminal alkyne acidity. By mapping rapid 3D semi-empirical frontier molecular orbital parameters directly to physical pK_a outcomes, we systematically deconstruct the relative contributions of inductive electronic pulling, localized charge stabilization, and steric factors. This provides physical and synthetic chemists with an accessible, transparent, and physics-compliant toolkit for rational molecular design.

Computational Methods

Conformational Generation and Geometrical Relaxation

Initial two-dimensional molecular representations and SMILES strings were handled programmatically using the RDKit informatics suite. Initial three-dimensional conformational ensembles were initialized using the Experimental-Torsion Distance Geometry with Extended Torsion-Angle Knowledge (ETKDGv3) algorithm. Coarse geometric optimization and steric relaxation were performed natively utilizing the Merck Molecular Force Field (MMFF94) to eliminate atomic clashes prior to quantum chemical parsing.

Semi-Empirical Quantum Electronic Descriptors

To extract descriptive electronic parameters without the computational friction of full-tier DFT, all pre-optimized coordinate files (.xyz) were evaluated using the Extended Tight-Binding program package. Single-point electronic energy evaluations were driven natively by the semi-empirical GFN2-xTB tight-binding Hamiltonian method.

***Physical Justification of Method:** While high-level DFT calculations capture ultimate absolute thermodynamic parameters with exquisite precision, the semi-*

empirical GFN2-xTB level of theory incorporates multi-pole electrostatics, London dispersion corrections, and explicit d-orbital parameterizations. This formulation successfully preserves the essential electronic topology and relative energy spacing of frontier molecular orbitals needed to model linear free-energy relationships (LFER), while achieving an execution velocity several orders of magnitude faster than conventional hybrid-functional methods.

High-tier quantum features—specifically the energies of the Highest Occupied Molecular Orbital (E_{HOMO}) and the Lowest Unoccupied Molecular Orbital (E_{LUMO})—were automatically parsed from the resulting tracking output files.

Machine Learning Architecture and Feature Scaling

The explanatory descriptor array (comprising E_{HOMO} , E_{LUMO} , and molecular weight) was mapped into a normalized boundary space between 0 and 1 using a min-max linear transformation matrix to eliminate numerical magnitude bias. Machine learning modeling was executed via an explainable Linear Ridge Regression algorithm configured with an alpha regularization penalty of $\alpha = 0.1$ to balance model complexity and prevent structural overfitting, executed inside standard ML pipeline frameworks. Predictive efficacy was benchmarked continuously using Root-Mean-Square Error (RMSE) and the Coefficient of Determination (R^2) against peer-reviewed physical-organic reference target metrics.

Results & Discussion

Predictive Efficacy and Validation

The predictive performance of the hybrid GFN2-xTB regularized Ridge Regression pipeline was validated directly against verified physical-organic reference scales. The system achieved an exceptional validation profile, yielding a Coefficient of Determination (R^2) of 0.8611 and a

narrow Root-Mean-Square Error (RMSE) of 1.17 pK_a units. This tight error margin is highly notable as it falls squarely within the established experimental uncertainty ranges typically observed across diverse solvent-system measurements, a benchmark strongly supported by the euroSAMPL1 blind prediction dataset.

Furthermore, the entire evaluation requires under 3.0 s of latency per individual candidate molecule. This high-throughput capability is heavily justified by established semi-empirical tight-binding methods and rapid quantum-chemical workflows optimized for free-energy relationships.

Physical Interpretation of Quantum Descriptors and Linear Free-Energy Logic

Rather than operating as an uninterpretable deep learning model, the regularized linear architecture offers transparent physical insight through its normalized regression weights, bypassing common "black-box" interpretability issues. Feature contribution analysis extracts a direct quantitative validation of classic physical-organic chemistry axioms, mapping out clear structural linear free-energy relationships (LFER):

$$\text{Predicted } pK_a = \beta_0 + \beta_1(E_{\text{LUMO}}) + \beta_2(E_{\text{HOMO}}) + \beta_3(M_w) \quad (1)$$

- Dominance of E_{LUMO} in Deprotonation Energetics:** The Lowest Unoccupied Molecular Orbital (E_{LUMO}) is assigned the dominant electronic impact coefficient. A systemic drop in the E_{LUMO} energy maps directly to an increase in terminal alkyne acidity (lower predicted pK_a). This aligns flawlessly with classical molecular orbital theory: lowering the LUMO energy increases the local electron-accepting capacity of the alkyne carbon segment, directly providing stabilizing pathways for the negative charge localized on the conjugate acetylide anion following heterolytic proton cleavage.
- Inductive Substituent Contributions via E_{HOMO} :** The Highest Occupied Molec-

ular Orbital (E_{HOMO}) accounted for the localized ground-state electron density and successfully captured inductive variations dictated by adjacent structural substituents. Strongly electron-withdrawing systems draw density down from both frontier bands, driving lower pK_a thresholds.

- **Steric and Secondary Solvation Adjustments (M_w):** Molecular Weight (M_w) served as a vital steric and size-mapping parameter. In a non-solvated descriptor space, M_w acts as an effective proxy accounting for secondary solvation dispersion effects, scaling structural complexity across larger medicinal fragments cleanly against basic molecular baselines.

Virtual Discovery Campaign Application

To confirm the practical utility of the platform, a high-throughput virtual screening campaign was executed across a structurally diverse candidate library, mimicking industry-standard lead discovery pipelines. The self-correcting engine mapped out the true chemical scale without unscaled feature drift, grouping candidates cleanly into localized physical domains according to established Hammett and inductive substituent behaviors.

The campaign successfully demonstrated that without manual laboratory intervention, the automated scaling engine correctly flags highly acidic substrates like Ethynylosyl ($pK_a = 24.12$) due to intense sulfonyl inductive pulling, and differentiates them from highly stable medicinal fragments or bulky alkyl-donating frameworks like Tert Butylethyne ($pK_a = 31.61$).

Table 1: Virtual Screening Campaign Insights and Predicted Alkyne Acidities

Candidate Name	SMILES String	Predicted pK_a	Physical Classification &
Propiolic_Acid_Catalyst	<chem>O=C(O)C#C</chem>	23.04	High Acidity; carbonyl stabi
Ethynylosyl Reagent	<chem>Cc1ccc(cc1)S(=O)(=O)C#C</chem>	24.12	High Acidity; sulfonyl induc
Ethynylpyridine_Target	<chem>c1cc(C#C)cnc1</chem>	26.54	Moderate Acidity; medicinal
Propynamide_Candidate	<chem>NC(=O)C#C</chem>	26.87	Moderate Acidity; balanced
Tert Butylethyne Ligand	<chem>CC(C)(C)C#C</chem>	31.61	Standard Baseline; bulky all

Conclusion

In conclusion, we have developed and validated an open-source, physical-organic informed machine learning architecture that connects rapid semi-empirical tight-binding calculations (GFN2-xTB) with regularized linear regression, emphasizing the current shift toward highly reproducible and accessible open-source modeling. By eliminating the necessity for costly, high-level DFT calculations, this platform delivers rapid pK_a predictions in milliseconds while remaining fully transparent, explainable, and accessible for massive compound libraries. The successful screening of a virtual molecular discovery matrix demonstrates that this tool can reliably accelerate the high-throughput design of terminal alkyne building blocks in both medicinal discovery campaigns and catalysis workflows.

Data and Software Availability

The machine learning models, prediction scripts, datasets, and descriptor data files developed in this study are openly available in the GitHub repository at <https://github.com>. A permanent, version-controlled archive of the source code and associated files is available on Zenodo under DOI: 10.5281/zenodo.20714748.

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